Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$\int \frac{\left(x^2-1\right)dx}{x\sqrt{(x^2+\alpha x+1)(x^2+\beta x+1)}}$ is equal to

(A) $\log\left[\frac{\sqrt{x^2+\alpha x+1}+\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right]+c$

(B) $2\log\left[\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right]+c$

(C) $\log\left[\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}\right]+c$

(D) None of these

Q.2

If $\int \tan^7 x dx = f(x)$ then

(A) $\mathbf{f(x)}$ is a polynomial of degree 8 in $\tan \mathbf{x}$

(B) $f(x)$ is a polynomial of degree 5 in $\tan x$

(C) $f(x)=\frac{1}{6}\tan^6 x-\frac{1}{4}\tan^4 x+\frac{1}{2}\tan^2 x+\log|\cos x|+C$

(D) $\mathrm{f(x)}$ is a polynomial of degree 6 in $\tan \mathrm{x}$

Q.3

$\int \frac{\log x^n}{x}dx, (n\in N)=$

(A) $\frac{1}{2n}(\log x)^2+C$

(B) $\frac{n}{2}(\log x)^2+C$

(C) $n(\log x)^2+C$

(D) None of these

Q.4

If $\int x^6 \sin\left(5x^7\right)dx=\frac{k}{5}\cos\left(5x^7\right), x\neq 0$, then

(A) $\mathbf{k=7}$

(B) $\mathbf{k=-7}$

(C) $\mathrm{k}=\frac{1}{7}$

@IITJEE_Advanced

(D) $\mathrm{k} = -\frac{1}{7}$

Q.5

$\int \frac{\sin 2x}{\mathrm{a}\cos^2 x + b\sin^2 x} dx =$

(A) $(\mathrm{b} - \mathrm{a}) \log\left(\mathrm{a} - \cos^2 x + b\sin^2 x\right)$

(B) $\frac{1}{b-a} \log\left(\mathrm{a}\cos^2 x + b\sin^2 x\right)$

(C) $\frac{1}{b-a} \log\left(\mathrm{a}\cos^2 x - b\sin^2 x\right)$

(D) None of these

Q.6

$\int \frac{x^2 \tan^{-1} x^3}{1+x^6} dx =$

(A) $\tan^{-1}\left(\mathrm{x}^3\right) + \mathrm{C}$

(B) $\frac{1}{6}\left(\tan^{-1} x^3\right)^2 + C$

(C) $\frac{1}{2}\left(\tan^{-1} x^3\right)^2 + C$

(D) $\frac{1}{2}\left(\tan^{-1} x^2\right)^3 + C$

Q.7

$\int \sqrt{\frac{\cos x - \cos^3 x}{1 - \cos^3 x}} dx =$

(A) $\frac{2}{3}\sin^{-1}\left(\cos^{3/2} x\right) + C$

(B) $\frac{3}{2}\sin^{-1}\left(\cos^{3/2} x\right) + C$

(C) $\frac{2}{3}\cos^{-1}\left(\cos^{3/2} x\right) + C$

(D) None of these

Q.8

$\int \frac{dx}{1+\sin x+\cos x} =$

(A) $\log\left|1 + \tan\frac{x}{2}\right| + C$

(B) $\log\left|1 + \sin x + \cos x\right| + C$

(C) $2\log\left|1 + \tan\frac{x}{2}\right| + C$

(D) $\frac{1}{2}\log\left|1 + \tan\frac{x}{2}\right| + C$

Q.9

@IITJEE_Advanced

Let the equation of a curve passing through the point $(0, 1)$ be given by $y = \int x^2 \cdot e^{x^3} dx$. If the equation of the curve is written in the form $x = f(y)$ then $f(y)$ is

(A) $\sqrt{\log_e(3y - 2)}$

(B) $\sqrt[3]{\log_e(3y - 2)}$

(C) $\sqrt[3]{\log_e(2 - 3y)}$

(D) None of these

Q.10

If $\int f(x) \sin x \cos x dx = \frac{1}{2(b^2 - a^2)} \log_e(f(x)) + A$
$b \neq \pm a$, then $\{f(x)\}^{-1}$ is equal to

(A) $a^2 \sin^2 x + b^2 \cos^2 x + C$

(B) $a^2 \sin^2 x - b^2 \cos^2 x + C$

(C) $a^2 \cos^2 x + b^2 \sin^2 x + C$

(D) $a^2 \cos^2 x - b^2 \sin^2 x + C$

Q.11

$\int \sqrt{\frac{x}{1-x}} dx$ is equal to

(A) $\sin^{-1} \sqrt{x} + c$

(B) $\sin^{-1}\{\sqrt{x} - \sqrt{x(1 - x)}\} + c$

(C) $\sin^{-1} \sqrt{x(1 - x)} + c$

(D) $\sin^{-1} \sqrt{x} - \sqrt{x(1 - x)} + c$

Q.12

If $\int \sqrt{1 + \sec x} dx = 2(fog)(x) + C$, then

(A) $f(x) = \sec x - 1$

(B) $f(x) = 2 \tan^{-1} x$

(C) $g(x) = \sqrt{\sec x - 1}$

(D) None of these

Q.13

If $f(x) = \ln\left(x - \sqrt{1 + x^2}\right)$, then what is $\int f''(x) dx$
equal to ?

@IITJEE_Advanced

(A) $\frac{1}{\left(x-\sqrt{1+x^2}\right)}+c$

(B) $-\frac{1}{\sqrt{1+x^2}}+c$

(C) $-\sqrt{1+x^2}+c$

(D) $\ln\left(x-\sqrt{1+x^2}\right)+c$

Q.14

If $\int g(x)dx=g(x)$, then
$\int g(x)\left\{f(x)+f'(x)\right\}dx$ is equal to

(A) $g(x)f(x)-g(x)f'(x)+C$

(B) $g(x)f'(x)+C$

(C) $g(x)f(x)+C$

(D) $g(x)f^2(x)+C$

Q.15

$\int \frac{dx}{\sin x(3+\cos^2 x)}$ is equal to

(A) $\log\left|y^2-1\right|-\tan^{-1}y+C$

(B) $\tan^{-1}\frac{y}{\sqrt{3}}+C$

(C) $\log\left|\frac{y-1}{y+1}\right|+C$

(D) $\frac{1}{4}\log\left|\frac{y-1}{y+1}\right|-\frac{1}{4\sqrt{3}}\tan^{-1}\frac{y}{\sqrt{3}}+C$

Q.16

If $\int \sec x\,\mathrm{cosec}\,x dx=\log|g(x)|+c$, then what
is $g(x)$ equal to ?

(A) $\sin x\cos x$

(B) $\sec^2 x$

(C) $\tan x$

(D) $\log|\tan x|$

Q.17

If $\int f(x)\cos x dx=\frac{1}{2}f^2(x)+c$, then $f(x)$ can be

(A) x

(B) 1

@IITJEE_Advanced

(C) $\cos x$

(D) $\sin x$

Q.18

$\int \frac{\{f(x)\cdot\phi'(x)-f'(x)\cdot\phi(x)\}}{f(x)\cdot\phi(x)} \log \frac{\phi(x)}{f(x)} dx$ is equal to :

(A) $\log \frac{\phi(x)}{f(x)} + k$

(B) $\frac{1}{2}\left\{\log \frac{\phi(x)}{f(x)}\right\}^2 + k$

(C) $\frac{\phi(x)}{f(x)} \log \frac{\phi(x)}{f(x)} + k$

(D) None of these

Q.19

$\int \frac{x^2-1}{(x+1)^2\sqrt{x^3+x^2+x}} dx$ is equal to

(A) $\tan^{-1}\sqrt{\frac{x^2+x+1}{x}} + C$

(B) $2\tan^{-1}\sqrt{\frac{x^2+x+1}{x}} + C$

(C) $3\tan^{-1}\sqrt{\frac{x^2+x+1}{x}} + C$

(D) None of these

Q.20

What is $\int \frac{\log x}{(1+\log x)^2} dx$ equal to?

(A) $\frac{1}{(1+\log x)^3} + c$

(B) $\frac{1}{(1+\log x)^2} + c$

(C) $\frac{x}{(1+\log x)} + c$

(D) $\frac{x}{(1+\log x)^2} + c$

Where c is a constant

Answers & Solutions

Q.1 Answer:

$2\log\left[\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right] + c$

Solution:

@IITJEE_Advanced

$$I=\int \frac{x^2\left(1-\frac{1}{x^2}\right)dx}{x^2\left[\sqrt{\left(x+\frac{1}{x}+a\right)}\sqrt{x+\frac{1}{x}+\beta}\right]}$$

$\text{Put } x+\frac{1}{x}=z \therefore \left(1-\frac{1}{x^2}\right)dx=dz$

$\Rightarrow I=\int \frac{dz}{\sqrt{(z+a)(z+\beta)}}$

$=\int \frac{dz}{\sqrt{z^2+(a+\beta)z+a\beta}}$

$=\int \frac{dz}{\sqrt{\left(z+\frac{a+\beta}{2}\right)^2-\left(\frac{a-\beta}{2}\right)^2}}$

$=\log\left|z+\frac{a+\beta}{2}-\sqrt{(z+a)(z+\beta)}\right|+c$

$=\log\left|x+\frac{1}{x}+\frac{a+\beta}{2}-\sqrt{\left(x+\frac{1}{x}+a\right)\left(x+\frac{1}{x}+\beta\right)}\right|+c$

$=\log\left[\frac{2x^2+(a+\beta)x+2}{2x}-\frac{\sqrt{(x^2+\alpha x+1)(x^2+\beta x+1)}}{x}\right]+c$

$=\log\frac{1}{2}\left(\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right)^2+c$

$=2\log\left(\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right)^2+c$

Q.2 Answer:

$f(x)=\frac{1}{6}\tan^6 x-\frac{1}{4}\tan^4 x+\frac{1}{2}\tan^2 x+\log|\cos x|+C$

Solution:

Putting $\tan x=t$, we have $dx=\frac{dt}{1+t^2}$ so

$\int \tan^7 x dx=\int \frac{t^7}{1+t^2}dt=\int\left(t^5-t^3+t-\frac{t}{1+t^2}\right)dt$

$=\frac{t^6}{6}-\frac{t^4}{4}+\frac{t^2}{2}-\frac{1}{2}\log\left(1+t^2\right)+C$

$=\frac{1}{6}\tan^6 x-\frac{1}{4}\tan^4 x+\frac{1}{2}\tan^2 x-\frac{1}{2}\log\sec^2 x+C=\frac{1}{6}\tan^6 x-\frac{1}{4}\tan^4 x+\frac{1}{2}\tan^2 x+\log|\cos x|+C$

Q.3 Answer:

$\frac{n}{2}(\log x)^2+C$

Solution:

$I=\int \frac{n\log x}{x}dx=n\int(\log x)\times\frac{1}{x}dx$

$=n\frac{(\log x)^2}{2}+C$

Q.4 Answer:

$k=-\frac{1}{7}$

Solution:

Put $5x^7=t\Rightarrow 35x^6dx=dt$

$\therefore \int x^6\sin\left(5x^7\right)dx=\int \sin t\frac{1}{35}dt=-\frac{1}{35}\cos t$

$=-\frac{1}{35}\cos\left(5x^7\right)\Rightarrow -\frac{1}{35}=\frac{k}{5}\Rightarrow k=-\frac{1}{7}$

@IITJEE_Advanced

Q.5 Answer:

$\frac{1}{b-a}\log\left(a\cos^2 x+b\sin^2 x\right)$

Solution:

Put $a\cos^2 x+b\sin^2 x=t$

$\Rightarrow (b-a)2\sin x\cos x dx=dt$

$\therefore \mathrm{I}=\frac{1}{\mathrm{b-a}}\int\frac{\mathrm{dt}}{\mathrm{t}}$, **etc.**

Q.6 Answer:

$\frac{1}{6}\left(\tan^{-1}x^3\right)^2+C$

Solution:

Put $\tan^{-1}x^3=t\Rightarrow\frac{1}{1+x^6}3x^2dx=dt$

$\therefore \mathrm{I}=\frac{1}{3}\int\mathrm{tdt}=\frac{1}{3}\frac{\mathrm{t}^2}{2}+\mathrm{C}$

Q.7 Answer:

$\frac{2}{3}\cos^{-1}\left(\cos^{3/2}x\right)+C$

Solution:

$I=\int\frac{\sqrt{\cos x(1-\cos^2 x)}}{\sqrt{1-\cos^3 x}}dx=\int\frac{\sqrt{\cos x}\sin x}{\sqrt{1-\left(\cos^{3/2}x\right)^2}}dx$

Put $\cos^{3/2}x=t\Rightarrow\frac{3}{2}\sqrt{\cos x}(-\sin x)dx=dt$

$I=\frac{2}{3}\int\frac{-dt}{\sqrt{1-t^2}}$

$=\frac{2}{3}\cos^{-1}(\mathrm{t})+\mathrm{C}=\frac{2}{3}\cos^{-1}\left(\cos^{3/2}\mathrm{x}\right)+\mathrm{C}$

Q.8 Answer:

$\log\left|1+\tan\frac{x}{2}\right|+C$

Solution:

$I=\int\frac{dx}{(1+\cos x)+\sin x}=\int\frac{dx}{2\cos^2\frac{x}{2}+2\sin\frac{x}{2}\cos\frac{x}{2}}$ **Divide Num. and Den. by** $\cos^2\frac{x}{2}$

$=\int\frac{\sec^2\frac{x}{2}\mathrm{dx}}{2\left(1+\tan\frac{x}{2}\right)}=\log\left|1+\tan\frac{x}{2}\right|+C$

Q.9 Answer:

$\sqrt[3]{\log_{\mathrm{e}}(3\mathrm{y}-2)}$

Solution:

$y=\int x^2\cdot e^{x^3}dx$

Put $x^3=t\Rightarrow 3x^2dx=dt$

$y=\int e^t\cdot\frac{dt}{3}=\frac{1}{3}e^t+C=\frac{1}{3}e^{x^3}+C$

It passes through $(0,1)$ **then** $1=\frac{1}{3}\mathrm{e}^0+\mathrm{C}\Rightarrow\mathrm{C}=1-\frac{1}{3}=\frac{2}{3}$

$\therefore y=\frac{1}{3}e^{x^3}+\frac{2}{3}$

@IITJEE_Advanced

$\Rightarrow e^{x^3} + 2 = 3y \Rightarrow e^{x^3} = 3y - 2$

$\Rightarrow x^3 = \log_e(3y - 2) \Rightarrow x = \sqrt[3]{\log_e(3y - 2)}$

Q.10 Answer:

$a^2 \sin^2 x + b^2 \cos^2 x + C$

Solution:

$\int f(x) \sin x \cos x dx = \frac{1}{2(b^2 - a^2)} \log(f(x)) + C$

therefore

$f(x) \sin x \cos x = \frac{1}{2(b^2 - a^2)} \cdot \frac{1}{f(x)} f'(x)$

[by differentiating both the sides] $\Rightarrow 2\left(b^2 - a^2\right) \sin x \cos x = \frac{f'(x)}{(f(x))^2}$

$\int \left(2b^2 \sin x \cos x - 2a^2 \sin x \cos x\right) dx = \int \frac{f'(x)}{(f(x))^2} dx$

[by integrating both the sides] $\Rightarrow -b^2 \cos^2 x - a^2 \sin^2 x - C = -\frac{1}{f(x)}$

Q.11 Answer:

$\sin^{-1} \sqrt{x} - \sqrt{x(1 - x)} + c$

Solution:

Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta$

$\int \sqrt{\frac{x}{1-x}} dx = \int \frac{\sin \theta}{\cos \theta} \cdot 2 \sin \theta \cos \theta d\theta$

$= \int (1 - \cos 2\theta) d\theta = \theta - \frac{1}{2} \sin 2\theta + c$

$= \sin^{-1} \sqrt{x} - \sqrt{x(1 - x)} + c$

Q.12 Answer:

$g(x) = \sqrt{\sec x - 1}$

Solution:

$\int \sqrt{1 + \sec x} dx = \int \frac{\sec x \cdot \tan x}{\sec x \cdot \sqrt{\sec x - 1}} dx = \int \frac{\tan x}{\sec x - 1} dx$

$(\text{Put } \sec x = t \Rightarrow \sec x \tan x dx = dt)$

$= \int \frac{dt}{t\sqrt{t - 1}} = 2 \int \frac{ydy}{y^2 + 1y}$ where $y^2 = t - 1$

$= 2\tan^{-1} y + C = 2\tan^{-1} \sqrt{\sec x - 1} + C$

$\therefore f(x) = \tan^{-1} x$ and $g(x) = \sqrt{\sec x - 1}$.

Q.13 Answer:

$-\frac{1}{\sqrt{1+x^2}} + c$

Solution:

Given that $f(x) = \ln\left(x - \sqrt{1 + x^2}\right)$

$\int f''(x) dx = f'(x) + c$ **where c is a constant**

@IITJEE_Advanced

$= \frac{1}{\left(x-\sqrt{1+x^2}\right)} \cdot \left(1-\frac{2x}{2\sqrt{1+x^2}}\right) + c$

$= \frac{-\left(x-\sqrt{1+x^2}\right)}{\left(\sqrt{1+x^2}\right)\left(x-\sqrt{1+x^2}\right)} + c = -\frac{1}{\sqrt{1+x^2}} + c$

Q.14 Answer:

$g(x)f(x) + C$

Solution:

$\int g(x)\{f(x)+f'(x)\}dx$

$= \int g(x)f(x)dx + \int g(x)f'(x)dx$

$= f(x)\int g(x)dx - \int \left\{f'(x)\int g(x)dx\right\}dx + \int g(x)f'(x)dx$

$\left[\text{ Given } \int g(x)dx = g(x)\right]$

$= f(x)g(x) + c$

Q.15 Answer:

$\frac{1}{4}\log\left|\frac{y-1}{y+1}\right| - \frac{1}{4\sqrt{3}}\tan^{-1}\frac{y}{\sqrt{3}} + C$

Solution:

$\int \frac{dx}{\sin x(3+\cos^2 x)}$

$= \int \frac{\sin x dx}{\sin^2 x(3+\cos^2 x)}$

$= \int \frac{\sin x dx}{(1-\cos^2 x)(3+\cos^2 x)}$

$= \int \frac{dy}{(y^2-1)(y^2+3)}(\text{ Putting } \cos x = y)$

$= \frac{1}{4}\int\left[\frac{1}{y^2-1} - \frac{1}{y^2+3}\right]dy$

$= \frac{1}{4}\log\left|\frac{y-1}{y+1}\right| - \frac{1}{4\sqrt{3}}\tan^{-1}\frac{y}{\sqrt{3}} + C$

Q.16 Answer:

$\tan x$

Solution:

Let $I = \int \sec x \cdot \operatorname{cosec} x dx$

$= \int \frac{2}{2\sin x\cos x}dx$

$= 2\int \frac{1}{\sin 2x}dx - 2\int \frac{1}{\frac{2\tan x}{1+\tan^2 x}}$

$\left[\because \sin 2x = \frac{2\tan x}{1+\tan^2 x}\right]$

$= \int \frac{\sec^2 x}{\tan x}dx$

Let $\tan x = t \Rightarrow \sec^2 dx = dt$

So, $\mathrm{I} = \int \frac{\mathrm{dt}}{\mathrm{t}} = \log|\mathrm{t}| + \mathrm{c} = \log|\tan \mathrm{x}| + \mathrm{c}$

But $\int \sec x \operatorname{cosec} x dx = \log|g(x)| + c$

$\therefore \quad g(x) = \tan x$

Q.17 Answer:

@IITJEE_Advanced

$\sin x$

Solution:

Since $\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c$
$\therefore \int \mathrm{f(x)\cos x dx} = \frac{\mathrm{f^2(x)}}{2} + \mathrm{c}$
$\Rightarrow \mathrm{f'(x) = \cos x \Rightarrow f(x) = \sin x}$

Q.18 Answer:

$\frac{1}{2}\left\{\log \frac{\phi(x)}{f(x)}\right\}^2 + k$

Solution:

$I = \int \frac{f(x)\phi'(x) - f'(x)\phi(x)}{f(x)\phi(x)} \log\left(\frac{\phi(x)}{f(x)}\right) dx$
Putting $\log \frac{\phi(\mathrm{x})}{\mathrm{f(x)}} = \mathrm{t}$
$\frac{f(x)}{\phi(x)} \cdot \frac{f(x)\phi'(x) - f'(x)\phi(x)}{(f(x))^2} \cdot dx = dt$, **we get**
$I = \int tdt = \frac{1}{2}t^2 + k = \frac{1}{2}\left(\log \frac{\phi(x)}{f(x)}\right)^2 + k, k \in \mathbf{R}$

Q.19 Answer:

$2\tan^{-1}\sqrt{\frac{x^2+x+1}{x}} + C$

Solution:

$\int \frac{x^2-1}{(x+1)^2\sqrt{x^3+x^2+x}}dx$
$= \int \frac{\left(\mathrm{x^2-1}\right)}{(\mathrm{x^2+2x+1})\cdot \mathrm{x}\sqrt{\mathrm{x}+\frac{1}{\mathrm{x}}+1}}\mathrm{dx}$
$= \int \frac{\left(1-\frac{1}{x^2}\right)}{\left(x+\frac{1}{x}+2\right)\sqrt{x+\frac{1}{x}+1}}dx = \int \frac{2zdz}{(z^2+1)\cdot z}$
[**Putting** $x + \frac{1}{x} + 1 = z^2 \Rightarrow \left(1 - \frac{1}{x^2}\right)dx = 2zdz$
$= 2\int \frac{\mathrm{dz}}{1+\mathrm{z^2}} = 2\tan^{-1}\mathrm{z} + \mathrm{C}$
$= 2\tan^{-1}\sqrt{\frac{\mathrm{x^2+x+1}}{\mathrm{x}}} + \mathrm{C}$

Q.20 Answer:

$\frac{x}{(1+\log x)} + c$

Solution:

Let $I = \int \frac{\log x}{(1+\log x)^2}dx$
Put $\log x = t \Rightarrow \frac{1}{x}dx = dt$
$I = \int \frac{e^t t}{(1+t)^2}dt = \int \frac{e^t\cdot(t+1-1)}{(1+t)^2}dt$
$= \int \frac{e^t(1+t)}{(1+t)^2}dt - \int \frac{e^t}{(1+t)^2}dt$
$= \int \frac{e^t}{1+t}dt - \int \frac{e^t}{(1+t)^2}dt$
$= \frac{e^t}{1+t} - \int -e^t\frac{1}{(1+t)^2}dt - \int \frac{e^t}{(1+t)^2}dt$

@IITJEE_Advanced

$= \frac{e^t}{1+t} + \int^{e^t} \frac{1}{(1+t)^2}\,dt - \int \frac{e^t}{(1+t)^2}\,dt$

$= \frac{e^t}{1+t} + c = \frac{x}{1+\log x} + c$

@IITJEE_Advanced